# Brain Tumor MRI Classification Using Deep Learning with Uncertainty Estimation

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## I. Introduction

Brain tumors are among the most dangerous medical conditions. Early and accurate detection directly affects treatment options and patient outcomes. Radiologists use MRI scans to identify tumor type, but manual review is time-consuming and can be inconsistent across different readers.

Artificial intelligence, and deep learning in particular, has shown strong performance in automating medical image analysis. However, most AI systems simply output a prediction without indicating how confident they are. In a clinical setting, an overconfident wrong prediction can lead to harmful treatment decisions. This paper addresses that gap by building a model that also reports its own uncertainty, allowing risky predictions to be forwarded for expert review.

Three approaches are compared: a Random Forest baseline (BML), an SVM with PCA reduction (AML), and a deep learning model based on ResNet-18 (DL). The DL model is the main contribution of this work.

## II. Dataset

The Brain Tumor MRI Dataset from Kaggle contains approximately 7,000 MRI images across four classes. The dataset is near-balanced, making accuracy a fair metric. All images are resized to 224×224 pixels.

**TABLE I. Dataset Class Distribution**

| Class | Count | Description |
|---|---|---|
| Glioma | ~1,621 | Malignant; irregular edges |
| Meningioma | ~1,645 | Benign; smooth boundary |
| No Tumor | ~2,000 | Healthy brain tissue |
| Pituitary | ~1,757 | Pituitary gland tumor |

## III. System Overview

This project builds three progressively more powerful classifiers. The Baseline ML (BML) is a Random Forest with 200 trees trained on hand-crafted image features: HOG edge descriptors, LBP texture patterns, GLCM co-occurrence statistics, and intensity statistics — about 370 features total.

The Advanced ML (AML) model uses a Support Vector Machine with an RBF kernel, applied after PCA reduces the 370 features down to around 100. Platt Scaling is used to convert SVM scores into calibrated probabilities.

The Deep Learning (DL) model, the main focus of this paper, uses ResNet-18 pre-trained on ImageNet and fine-tuned on the MRI dataset. It learns its own features automatically and uses MC-Dropout for uncertainty estimation.

## IV. Deep Learning Model

### A. ResNet-18 Architecture

ResNet-18 is a 18-layer convolutional neural network originally trained on 1.2 million ImageNet images. Its key innovation is the use of skip connections: each residual block adds its input directly to its output. This helps gradients flow backward during training and allows the network to be much deeper without losing performance.

For this task, the first two residual blocks are frozen — their weights are kept as-is from ImageNet training. Only the deeper layers and the final classification head are updated. This is called transfer learning: the model reuses general visual knowledge (edges, textures) and adapts the higher-level reasoning to brain tumors.

The final fully connected layer is replaced with a Dropout layer (rate 0.3) followed by a linear layer mapping to 4 output classes. The Dropout layer serves double duty: it regularizes training to reduce overfitting, and it enables MC-Dropout uncertainty estimation at inference time.

### B. Training Configuration

Training uses the AdamW optimizer (learning rate 0.001, weight decay 0.0001) with cosine annealing to reduce the learning rate smoothly over 5 epochs. Label smoothing of 0.1 is applied to the cross-entropy loss to prevent the model from being overconfident. Early stopping saves the best checkpoint based on validation accuracy.

Data augmentation is applied to every training image: random horizontal flips (50% chance), rotations up to 15 degrees, small translations and scale changes, and random brightness/contrast adjustments. These simulate real-world scan variation due to patient positioning and scanner differences.

## V. Uncertainty Estimation

### A. The Problem with Standard Predictions

A standard neural network runs a single forward pass and outputs a set of class probabilities. These probabilities can look confident even when the model is actually uncertain or wrong. For medical decisions, this is dangerous — a confident wrong prediction may not be questioned.

### B. MC-Dropout Method

MC-Dropout keeps the Dropout layer active during prediction instead of turning it off as usual. The same image is passed through the network 20 times. Each pass randomly deactivates different neurons, producing slightly different predictions. By averaging across all 20 passes, we get a more reliable prediction, and by measuring how much the 20 runs disagreed, we estimate uncertainty.

Two uncertainty measures are produced. Entropy measures how evenly spread the final probabilities are — if the model is very sure about one class, entropy is low; if it is split between classes, entropy is high. Prediction Standard Deviation measures the spread of the 20 individual predictions around the average. Both measures are higher for wrong predictions than for correct ones, validating their use as safety signals.

Any prediction with a maximum class probability below 0.70 is flagged automatically as low-confidence and recommended for radiologist review.

## VI. Results

### A. Overall Performance

**TABLE II. Model Performance Comparison**

| Model | Acc. | Macro F1 | Brier |
|---|---|---|---|
| RF (BML) | 85.0% | 0.83 | 0.08 |
| SVM (AML) | 87.0% | 0.85 | 0.09 |
| ResNet (DL) | 97.98% | 0.98 | 0.01 |

The DL model outperforms both baselines by a large margin. The Brier score of 0.01 means the model's stated confidence closely matches its actual correctness — a property called calibration that is critical for trustworthy medical AI.

### B. Per-Class Results

**TABLE III. ResNet-18 Per-Class Metrics**

| Class | Precision | Recall | F1 |
|---|---|---|---|
| Glioma | 0.98 | 0.98 | 0.98 |
| Meningioma | 0.97 | 0.97 | 0.97 |
| No Tumor | 0.99 | 0.99 | 0.99 |
| Pituitary | 0.99 | 0.98 | 0.99 |

Meningioma, historically the hardest class due to its visual overlap with Glioma, improves from F1 of 0.81 in BML to 0.97 in the DL model. All four classes exceed 0.97 F1, demonstrating consistent and reliable performance.

### C. Why Deep Learning Outperforms Traditional ML

**TABLE IV. Comparison of Approaches**

| Aspect | BML / AML | DL (ResNet-18) |
|---|---|---|
| Features | Hand-crafted | Learned from data |
| Boundaries | Linear / RBF | 18 non-linear layers |
| Calibration | Platt Scaling | MC-Dropout ensemble |
| Accuracy | 85–87% | 97.98% |
| Uncertainty | Basic entropy | True variance + entropy |

The fundamental advantage of deep learning is that it learns its own features end-to-end, optimized for the exact task. Traditional ML uses fixed feature extractors designed by humans, which cannot adapt to subtle patterns in MRI scans. Transfer learning further boosts performance by providing a rich starting point from ImageNet training.

## VII. Discussion

The most clinically significant result is that the model knows when it does not know. Wrong predictions consistently show higher entropy and higher prediction variance. This means a simple threshold on the confidence score can identify most errors before they reach a clinician.

One practical limitation is the inference time cost of 20 forward passes. In a real-time clinical setting, this could be reduced to 10 passes with minimal impact on uncertainty quality. Another consideration is dataset diversity — the model should be validated on scans from multiple hospitals and scanner types before deployment.

## VIII. Conclusion

This paper shows that ResNet-18 with MC-Dropout achieves 97.98% accuracy on brain tumor MRI classification while providing reliable uncertainty estimates. The system outperforms both the Random Forest baseline and the SVM advanced model, and includes a built-in safety mechanism that flags uncertain predictions for expert review. These properties make it a practical tool for AI-assisted radiology.

## References


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